Trigonometric identities

1 $\dfrac{\sin x}{\cos x}$

2 0

3

a $\dfrac{1}{2}$

b $\dfrac{a}{c}$

4

a $\dfrac{8}{17}$

b $-\dfrac{12}{13}$

5

a $\dfrac{4}{3}$

b $\dfrac{2}{\sqrt{5}}$

6

a $\sin\theta$

b $\sin y$

c $\tan\theta - 1$

7

a 2

b $\dfrac{17}{31}$

c $\dfrac{11}{6}$

d $\pm\dfrac{1}{\sqrt{2}}$

Pythagorean identities

8

a Yes

b No

9

a 1

b 4

10

a $-\sin^2(\theta)$

b $1-2\cos\theta\sin\theta$

c $-\cos^3(\theta)$

d $10-6\cos x$

e $\tan^2(\theta)$

f $\cos^2(\theta)$

11 $\dfrac{x^2}{16}+\dfrac{y^2}{9}=1$

Proofs

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- **a** No solution provided.
- **b** No solution provided.
- **c** No solution provided.
- **d** No solution provided.
- **e** No solution provided.
- **f** No solution provided.
- **g** No solution provided.
- **h** No solution provided.
- **i** No solution provided.
- **j** No solution provided.

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- **a** No solution provided.
- **b** No solution provided.
- **c** No solution provided.